TechSmart

3.3 Animating Shapes (1)

Coding Technique Practice

Technique1Practice1.py

```python
"""
LESSON: 3.3 - Animation
TECHNIQUE 1: Animating Shapes
PRACTICE 1
"""

import pygame
pygame.init()

window = pygame.display.set_mode([400, 400])
window.fill((200, 200, 255))

string_color = (240, 240, 240)
yoyo_color = (140, 40, 50)

string_start = (0, 0)
yoyo_x = 200
yoyo_y = 0

# Declare an offset for the yoyo


frames = 0
while frames < 100:

    # Add to the offset


    # Use yoyo_x, yoyo_y, and yoyo_offset in a point


    window.fill((0, 0, 0))

    # Draw a line, then a circle with a radius of
    # your choice to make a yoyo


```

```

    # Flip and wait every frame
    pygame.display.flip()
    pygame.time.wait(40)

    frames += 1

```

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3.3 Animating Shapes (1)

Coding Technique Practice

Technique1Practice1.py

```python
"""
LESSON: 3.3 - Animation
TECHNIQUE 1: Animating Shapes
PRACTICE 1
"""

import pygame
pygame.init()

window = pygame.display.set_mode([400, 400])
window.fill((200, 200, 255))

string_color = (240, 240, 240)
yoyo_color = (140, 40, 50)

string_start = (0, 0)
yoyo_x = 200
yoyo_y = 0

# Declare an offset for the yoyo
yoyo_offset = 0

frames = 0
while frames < 100:

    # Add to the offset
    yoyo_offset += 4

    # Use yoyo_x, yoyo_y, and yoyo_offset in a point
    yoyo_center = (yoyo_x, yoyo_y + yoyo_offset)

    window.fill((0, 0, 0))

    # Draw a line, then a circle with a radius of
    # your choice to make a yoyo
    pygame.draw.line(window, string_color, string_start, yoyo_center)
    pygame.draw.circle(window, yoyo_color, yoyo_center, 50)
```

```

    # Flip and wait every frame
    pygame.display.flip()
    pygame.time.wait(40)

    frames += 1

```

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3.3 Animating Shapes (2)

Coding Technique Practice

Technique1Practice2.py

```python
"""
LESSON: 3.3 - Animation
TECHNIQUE 1: Animating Shapes
PRACTICE 2
"""

import pygame
pygame.init()

window = pygame.display.set_mode([400, 400])
window.fill((200, 200, 255))
color = (100, 130, 80)

left = 0
right = 400
top = 0
bottom = 400

# Declare an offset for the corner


frames = 0
while frames <= 80:

    # Add to the offset


    # Use the side variables to make four corners,
    # adding or subtracting the offset from each




    window.fill((0, 0, 0))

    # Use the four points to draw a polygon

```

```

    # Flip and wait every frame


    frames += 1

```

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3.3 Animating Shapes (2)

Coding Technique Practice

Technique1Practice2.py

```python
"""
LESSON: 3.3 - Animation
TECHNIQUE 1: Animating Shapes
PRACTICE 2
"""

import pygame
pygame.init()

window = pygame.display.set_mode([400, 400])
window.fill((200, 200, 255))
color = (100, 130, 80)

left = 0
right = 400
top = 0
bottom = 400

# Declare an offset for the corner
offset = 0

frames = 0
while frames <= 80:

    # Add to the offset
    offset += 5

    # Use the side variables to make four corners,
    # adding or subtracting the offset from each
    left_upper = (left + offset, top)
    right_upper = (right, top + offset)
    right_lower = (right - offset, bottom)
    left_lower = (left, bottom - offset)

    window.fill((0, 0, 0))

    # Use the four points to draw a polygon
    pygame.draw.polygon(window, color, [left_upper, right_upper, right_lower, left_lower])
```

```

    # Flip and wait every frame
    pygame.display.flip()
    pygame.time.wait(40)
    frames += 1

```

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3.3 Animating Rect Shapes (1)

Coding Technique Practice

Technique2Practice1.py

```python
"""
LESSON: 3.3 - Animation
TECHNIQUE 2: Animating Rect Shapes
PRACTICE 1
"""

import pygame
pygame.init()

window = pygame.display.set_mode([400, 400])

sky_color = (180, 200, 240)
ground_color = (100, 150, 100)
building_color = (50, 50, 50)
elevator_color = (100, 100, 100)

ground_rect = pygame.Rect(0, 300, 400, 100)
building_rect = pygame.Rect(250, 20, 100, 280)

elevator_start = 240
elevator_rect = pygame.Rect(240, elevator_start, 40, 60)

# Declare an offset for the elevator


frames = 0
while frames < 100:

    # Subtract from offset


    # Use the offset to set the elevator rect's y


    window.fill(sky_color)
    pygame.draw.rect(window, ground_color, ground_rect)
```

```
    pygame.draw.rect(window, building_color, building_rect)

    # Draw the elevator


    # Flip and wait every frame



    frames += 1
```

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3.3 Animating Rect Shapes (1)

Coding Technique Practice

Technique2Practice1.py

```python
"""
LESSON: 3.3 - Animation
TECHNIQUE 2: Animating Rect Shapes
PRACTICE 1
"""

import pygame
pygame.init()

window = pygame.display.set_mode([400, 400])

sky_color = (180, 200, 240)
ground_color = (100, 150, 100)
building_color = (50, 50, 50)
elevator_color = (100, 100, 100)

ground_rect = pygame.Rect(0, 300, 400, 100)
building_rect = pygame.Rect(250, 20, 100, 280)

elevator_start = 240
elevator_rect = pygame.Rect(240, elevator_start, 40, 60)

# Declare an offset for the elevator
elevator_offset = 0

frames = 0
while frames < 100:

    # Subtract from offset
    elevator_offset -= 2

    # Use the offset to set the elevator rect's y
    elevator_rect.y = elevator_start + elevator_offset

    window.fill(sky_color)
    pygame.draw.rect(window, ground_color, ground_rect)
```

```
    pygame.draw.rect(window, building_color, building_rect)

    # Draw the elevator
    pygame.draw.rect(window, elevator_color, elevator_rect)

    # Flip and wait every frame
    pygame.display.flip()
    pygame.time.wait(40)

    frames += 1
```

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3.3 Animating Rect Shapes (2)

Coding Technique Practice

Technique2Practice2.py

```python
"""
LESSON: 3.3 - Animation
TECHNIQUE 2: Animating Rect Shapes
PRACTICE 2
"""

import pygame
pygame.init()

window = pygame.display.set_mode([400, 400])
window.fill((200, 200, 255))

bg_color = (120, 80, 40)
rock_color = (70, 60, 50)
ground_color = (60, 50, 40)
person_color = (150, 140, 80)

rock_y = -200
person_x = 50

ground_rect = pygame.Rect(0, 340, 400, 60)
rock_rect = pygame.Rect(20, rock_y, 150, 340)
person_rect = pygame.Rect(person_x, 300, 30, 40)

# Declare an offset for the rock and person



frames = 0
while frames < 100:

    # Increase the rock offset


    # If the rock offset is higher than 80,
    # increase the person offset



```

```
    # Use the offsets to set the rock rect's y
    # and person rect's x



    window.fill(bg_color)
    pygame.draw.rect(window, ground_color, ground_rect)

    # Draw the rock rectangle and person ellipse



    # Flip and wait every frame



    frames += 1
```

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3.3 Animating Rect Shapes (2)

Coding Technique Practice

Technique2Practice2.py

```python
"""
LESSON: 3.3 - Animation
TECHNIQUE 2: Animating Rect Shapes
PRACTICE 2
"""

import pygame
pygame.init()

window = pygame.display.set_mode([400, 400])
window.fill((200, 200, 255))

bg_color = (120, 80, 40)
rock_color = (70, 60, 50)
ground_color = (60, 50, 40)
person_color = (150, 140, 80)

rock_y = -200
person_x = 50

ground_rect = pygame.Rect(0, 340, 400, 60)
rock_rect = pygame.Rect(20, rock_y, 150, 340)
person_rect = pygame.Rect(person_x, 300, 30, 40)

# Declare an offset for the rock and person
rock_offset = 0
person_offset = 0

frames = 0
while frames < 100:

    # Increase the rock offset
    rock_offset += 2

    # If the rock offset is higher than 80,
    # increase the person offset
    if rock_offset > 80:
        person_offset += 3

```

```
    # Use the offsets to set the rock rect's y
    # and person rect's x
    rock_rect.y = rock_y + rock_offset
    person_rect.x = person_x + person_offset

    window.fill(bg_color)
    pygame.draw.rect(window, ground_color, ground_rect)

    # Draw the rock rectangle and person ellipse
    pygame.draw.rect(window, rock_color, rock_rect)
    pygame.draw.ellipse(window, person_color, person_rect)

    # Flip and wait every frame
    pygame.display.flip()
    pygame.time.wait(40)

    frames += 1
```

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3.3 Adjust Animation Speed (1)

Coding Technique Practice

Technique3Practice1.py

```python
"""
LESSON: 3.3 - Animation
TECHNIQUE 3: Adjust Animation Speed
PRACTICE 1
"""

import pygame
pygame.init()

window = pygame.display.set_mode([600, 400])

grass_color = (120, 240, 100)
car_color = (160, 20, 20)
road_color = (55, 50, 50)

car_x = 5

car_rect = pygame.Rect(car_x, 150, 50, 30)
road_rect = pygame.Rect(0, 120, 600, 160)

car_offset = 0

# Declare variable for speed


frames = 0
while frames < 100:

    # Change offsets by using speed
    car_offset += 1

    car_rect.x = car_x + car_offset

    window.fill(grass_color)
    pygame.draw.rect(window, road_color, road_rect)
    pygame.draw.ellipse(window, car_color, car_rect)

    pygame.display.flip()
    pygame.time.wait(40)
```

```

    frames += 1

    # Speed up car every five frames


```

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3.3 Adjust Animation Speed (1)

Coding Technique Practice

Technique3Practice1.py

```python
"""
LESSON: 3.3 - Animation
TECHNIQUE 3: Adjust Animation Speed
PRACTICE 1
"""

import pygame
pygame.init()

window = pygame.display.set_mode([600, 400])

grass_color = (120, 240, 100)
car_color = (160, 20, 20)
road_color = (55, 50, 50)

car_x = 5

car_rect = pygame.Rect(car_x, 150, 50, 30)
road_rect = pygame.Rect(0, 120, 600, 160)

car_offset = 0

# Declare variable for speed
car_speed = 1

frames = 0
while frames < 100:

    # Change offsets by using speed
    car_offset += car_speed

    car_rect.x = car_x + car_offset

    window.fill(grass_color)
    pygame.draw.rect(window, road_color, road_rect)
    pygame.draw.ellipse(window, car_color, car_rect)

    pygame.display.flip()
    pygame.time.wait(40)
```

```

    frames += 1

    # Speed up car every five frames
    if frames % 5 == 0:
        car_speed += 2
```

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3.3 Adjust Animation Speed (2)

Coding Technique Practice

Technique3Practice2.py

```python
"""
LESSON: 3.3 - Animation
TECHNIQUE 3: Adjust Animation Speed
PRACTICE 2
"""

import pygame
pygame.init()

window = pygame.display.set_mode([600, 400])

bg_color = (80, 80, 80)
black = (0, 0, 0)

jumper_y = 260
arrow_x = 0

jumper_rect = pygame.Rect(310, jumper_y, 60, 90)
arrow_rect = pygame.Rect(arrow_x, 270, 20, 10)
floor_rect = pygame.Rect(0, 350, 600, 50)
ceiling_rect = pygame.Rect(0, 0, 600, 50)

jumper_offset = 0
arrow_offset = 0
jump_speed = 0

# Declare variable for frame delay
Ω

frames = 0
while frames < 100:

    arrow_offset += 8
    jumper_offset += jump_speed

    arrow_rect.x = arrow_x + arrow_offset
    jumper_rect.y = jumper_y + jumper_offset

    window.fill(bg_color)
```

```
    pygame.draw.rect(window, black, floor_rect)
    pygame.draw.rect(window, black, ceiling_rect)
    pygame.draw.rect(window, black, arrow_rect)
    pygame.draw.rect(window, black, jumper_rect)

    pygame.display.flip()

    # Change to use the frame delay variable
    pygame.time.wait(40)

    frames += 1

    if frames == 20:
        jump_speed = -6
    if frames == 40:
        jump_speed = 0
    if frames == 45:
        jump_speed = 8
    if frames == 60:
        jump_speed = 0

    # Turn on slow motion on frame 30



    # Turn off slow motion on frame 50



```

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3.3 Adjust Animation Speed (2)

Coding Technique Practice

Technique3Practice2.py

```python
"""
LESSON: 3.3 - Animation
TECHNIQUE 3: Adjust Animation Speed
PRACTICE 2
"""

import pygame
pygame.init()

window = pygame.display.set_mode([600, 400])

bg_color = (80, 80, 80)
black = (0, 0, 0)

jumper_y = 260
arrow_x = 0

jumper_rect = pygame.Rect(310, jumper_y, 60, 90)
arrow_rect = pygame.Rect(arrow_x, 270, 20, 10)
floor_rect = pygame.Rect(0, 350, 600, 50)
ceiling_rect = pygame.Rect(0, 0, 600, 50)

jumper_offset = 0
arrow_offset = 0
jump_speed = 0

# Declare variable for frame delay
frame_delay = 40

frames = 0
while frames < 100:

    arrow_offset += 8
    jumper_offset += jump_speed

    arrow_rect.x = arrow_x + arrow_offset
    jumper_rect.y = jumper_y + jumper_offset

    window.fill(bg_color)
```

```
    pygame.draw.rect(window, black, floor_rect)
    pygame.draw.rect(window, black, ceiling_rect)
    pygame.draw.rect(window, black, arrow_rect)
    pygame.draw.rect(window, black, jumper_rect)

    pygame.display.flip()

    # Change to use the frame delay variable
    pygame.time.wait(frame_delay)

    frames += 1

    if frames == 20:
        jump_speed = -6
    if frames == 40:
        jump_speed = 0
    if frames == 45:
        jump_speed = 8
    if frames == 60:
        jump_speed = 0

    # Turn on slow motion on frame 30
    if frames == 30:
        frame_delay = 200

    # Turn off slow motion on frame 50
    if frames == 50:
        frame_delay = 40


```

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3.3 Backgrounds and Trails (1)

Coding Technique Practice

Technique4Practice1.py

```python
"""
LESSON: 3.3 - Animation
TECHNIQUE 4: Backgrounds and Trails
PRACTICE 1
"""

import pygame
pygame.init()

window = pygame.display.set_mode([400, 400])

bg_color = (0, 0, 0)

offset = 0

# Ask the user whether they want trails,
# and store the answer as a boolean



frames = 0
while frames < 100:

    offset += 1

    # Prevent trails if the user doesn't want trails



    circles = 11
    while circles > 0:
        circles -= 1
        pygame.draw.circle(window, (200, 0, circles * 20), (offset * circles, circles * 40), 25)

    pygame.display.flip()
    pygame.time.wait(40)

```

```
    frames += 1
```

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3.3 Backgrounds and Trails (1)

Coding Technique Practice

Technique4Practice1.py

```python
"""
LESSON: 3.3 - Animation
TECHNIQUE 4: Backgrounds and Trails
PRACTICE 1
"""

import pygame
pygame.init()

window = pygame.display.set_mode([400, 400])

bg_color = (0, 0, 0)

offset = 0

# Ask the user whether they want trails,
# and store the answer as a boolean
answer = input("Do you want the circles to leave trails? ")
trails = answer == "yes"

frames = 0
while frames < 100:

    offset += 1

    # Prevent trails if the user doesn't want trails
    if not trails:
        window.fill(bg_color)

    circles = 11
    while circles > 0:
        circles -= 1
        pygame.draw.circle(window, (200, 0, circles * 20), (offset * circles, circles * 40), 25)

    pygame.display.flip()
    pygame.time.wait(40)

```

```
    frames += 1
```

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3.3 Backgrounds and Trails (2)

Coding Technique Practice

Technique4Practice2.py

```python
"""
LESSON: 3.3 - Animation
TECHNIQUE 4: Backgrounds and Trails
PRACTICE 2
"""

import pygame
import random
pygame.init()

window = pygame.display.set_mode([400, 400])

bg_color = (0, 0, 0)

x_offset = 0
y_offset = 0

frames = 0
while frames < 110:

    y_offset += 25
    if y_offset > 400:
        x_offset += 80
        y_offset = 0

    # What do you change to make trails?
    window.fill(bg_color)

    # Make a random color, and change the circle
    # to use it

    pygame.draw.circle(window, color, (x_offset, y_offset), 50)

    pygame.display.flip()
    pygame.time.wait(40)

    frames += 1
```

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3.3 Backgrounds and Trails (2)

Coding Technique Practice

Technique4Practice2.py

```python
"""
LESSON: 3.3 - Animation
TECHNIQUE 4: Backgrounds and Trails
PRACTICE 2
"""

import pygame
import random
pygame.init()

window = pygame.display.set_mode([400, 400])

bg_color = (0, 0, 0)

x_offset = 0
y_offset = 0

frames = 0
while frames < 110:

    y_offset += 25
    if y_offset > 400:
        x_offset += 80
        y_offset = 0

    # What do you change to make trails?


    # Make a random color, and change the circle
    # to use it
    color = (random.randint(200, 255), random.randint(200, 255), random.randint(200, 255))
    pygame.draw.circle(window, color, (x_offset, y_offset), 50)

    pygame.display.flip()
    pygame.time.wait(40)

    frames += 1
```

TechSmart

3.3 Stick Dance: Random

Code Writing

StickDanceRandom_solution.py

```
"
LESSON: 3.3 - Animation
EXERCISE: Stick Dance (Random)
"""

#### ---- SETUP ---- ####

# Import the random library


# Import the PYGAME library

# INITialize pygame

# Create a new window with SET_MODE of size [300, 500],
# and assign to variable window




# Create a variable called frames with value 0


#### ---- ANIMATION LOOP ---- ####

# While frames is less than 100

    # Increment frames by 1
    # ---> TEST AFTER THIS LINE <--- #


    # FILL the window with WHITE


    #### ---- BODY ---- ####

```

```

    # Draw a CIRCLE on the window in BLACK with the
    # center point (150, 75), radius 50, and width 1


    # Draw a LINE on the window in BLACK with points
    # (150, 125) and (150, 275)
    # ---> TEST AFTER THIS LINE <--- #



    #### ---- LEGS ---- ####

    # Get a random integer from 25 to 100 and assign it
    # to variable left_leg


    # Get a random integer from 200 to 275 and assign
    # it to variable right_leg


    # Draw BLACK LINE from (150, 275) to (left_leg, 475)


    # Draw BLACK LINE from (150, 275) to (right_leg, 475)
    # ---> TEST AFTER THIS LINE <--- #



    #### ---- ARMS ---- ####

    # Get a random int between 100 and 250 and assign
    # to variable left_arm


    # Get another random int between 100 and 250 and
    # assign to variable right_arm


```

```
    # Draw BLACK LINE from (25, left_arm) to (150, 175)



    # Draw BLACK LINE from (275, right_arm) to
    # (150, 175)
    # ---> TEST AFTER THIS LINE <--- #



    #### ---- FINISH ---- ####

    # FLIP the display


    # WAIT for 100 milliseconds
    # ---> TEST AFTER THIS LINE <--- #



# Turn in your Coding Exercise.

```

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3.3 Stick Dance: Random

Code Writing

StickDanceRandom_solution.py

```python
"""
LESSON: 3.3 - Animation
EXERCISE: Stick Dance (Random)
"""

#### ---- SETUP ---- ####

# Import the random library
import random

# Import the PYGAME library
import pygame

# INITialize pygame
pygame.init()

# Create a new window with SET_MODE of size [300, 500],
# and assign to variable window
window = pygame.display.set_mode([300, 500])

# Create a variable called frames with value 0
frames = 0


#### ---- ANIMATION LOOP ---- ####

# While frames is less than 100
while frames < 100:

    # Increment frames by 1
    # ---> TEST AFTER THIS LINE <--- #
    frames += 1

    # FILL the window with WHITE
    window.fill((255, 255, 255))


    #### ---- BODY ---- ####
```

```

    # Draw a CIRCLE on the window in BLACK with the
    # center point (150, 75), radius 50, and width 1
    pygame.draw.circle(window, (0, 0, 0), (150, 75), 50, 1)

    # Draw a LINE on the window in BLACK with points
    # (150, 125) and (150, 275)
    # ---> TEST AFTER THIS LINE <--- #
    pygame.draw.line(window, (0, 0, 0), (150, 125), (150, 275))


    #### ---- LEGS ---- ####

    # Get a random integer from 25 to 100 and assign it
    # to variable left_leg
    left_leg = random.randint(25, 100)

    # Get a random integer from 200 to 275 and assign
    # it to variable right_leg
    right_leg = random.randint(200, 275)

    # Draw BLACK LINE from (150, 275) to (left_leg, 475)
    pygame.draw.line(window, (0, 0, 0), (150, 275), (left_leg, 475))

    # Draw BLACK LINE from (150, 275) to (right_leg, 475)
    # ---> TEST AFTER THIS LINE <--- #
    pygame.draw.line(window, (0, 0, 0), (150, 275), (right_leg, 475))


    #### ---- ARMS ---- ####

    # Get a random int between 100 and 250 and assign
    # to variable left_arm
    left_arm = random.randint(100, 250)

    # Get another random int between 100 and 250 and
    # assign to variable right_arm
    right_arm = random.randint(100, 250)

```

```
    # Draw BLACK LINE from (25, left_arm) to (150, 175)
    pygame.draw.line(window, (0, 0, 0), (25, left_arm), (150, 175))

    # Draw BLACK LINE from (275, right_arm) to
    # (150, 175)
    # ---> TEST AFTER THIS LINE <--- #
    pygame.draw.line(window, (0, 0, 0), (275, right_arm), (150, 175))


    #### ---- FINISH ---- ####

    # FLIP the display
    pygame.display.flip()

    # WAIT for 100 milliseconds
    # ---> TEST AFTER THIS LINE <--- #
    pygame.time.wait(100)



# Turn in your Coding Exercise.

```

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3.3 Healthful UFO

Code Restructure

HealthfulUFO_solution.py

```python
"""
LESSON: 3.3 - Animation
EXERCISE: Healthful UFO
"""

# Draw package
# Finish drawing
# Start the animating objects where they begin on screen
# Reset ship if off-screen
# Draw background
# Move package
# Move ship
# Make a package using a rectangle
# Reset package if ship is halfway across (x is 325)
# Draw ship
# Make a spaceship using two ellipses
# Animation variables


#### ---- SETUP ---- ####

import pygame
pygame.init()
window = pygame.display.set_mode([800, 500])


#### ---- ANIMATION VARIABLES ---- ####

# 1

ship_saucer = pygame.Rect(0, 10, 150, 40)
ship_cabin = pygame.Rect(40, 0, 70, 60)
ship_color = (150, 150, 150)
saucer_y = 130
cabin_y = 120
saucer_start_x = -150
cabin_start_x = -110
ship_speed = 5
```

```

# 2

package = pygame.Rect(0, 0, 30, 30)
package_x = 385
package_start_y = 145
package_speed = 7

# 3


ship_saucer.y = saucer_y
ship_cabin.y = cabin_y
package.x = package_x
package.y = package_start_y

# 4

ship_offset = 0
package_offset = 510  # the package starts off-screen the first time
packages_delivered = 0


#### ---- ANIMATION LOOP ---- ####

while packages_delivered < 3:

    #### ---- UPDATE POSITIONS ---- ####

    # 5

    ship_offset += ship_speed
    ship_saucer.x = saucer_start_x + ship_offset
    ship_cabin.x = cabin_start_x + ship_offset

    # 6

    package_offset += package_speed
    package.y = package_start_y + package_offset

    # 7

    if ship_saucer.x >= 950:
        ship_offset = 0
        packages_delivered += 1
```

```python


    # 8


    if ship_saucer.x == 325:
        package_offset = 0


    #### ---- DRAW ---- ####

    # 9

    window.fill((15, 5, 70))
    pygame.draw.circle(window, (45, 185, 225), (400, 900), 500)

    # 10

    pygame.draw.rect(window, (50, 175, 35), package)

    # 11

    pygame.draw.ellipse(window, ship_color, ship_saucer)
    pygame.draw.ellipse(window, ship_color, ship_cabin)

    # 12

    pygame.display.flip()
    pygame.time.wait(15)



# Turn in your Coding Exercise.


```

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3.3 Healthful UFO

Code Restructure

HealthfulUFO_solution.py

```python
"""
LESSON: 3.3 - Animation
EXERCISE: Healthful UFO
"""

# Draw package
# Finish drawing
# Start the animating objects where they begin on screen
# Reset ship if off-screen
# Draw background
# Move package
# Move ship
# Make a package using a rectangle
# Reset package if ship is halfway across (x is 325)
# Draw ship
# Make a spaceship using two ellipses
# Animation variables


#### ---- SETUP ---- ####

import pygame
pygame.init()
window = pygame.display.set_mode([800, 500])


#### ---- ANIMATION VARIABLES ---- ####

# 1
# Make a spaceship using two ellipses
ship_saucer = pygame.Rect(0, 10, 150, 40)
ship_cabin = pygame.Rect(40, 0, 70, 60)
ship_color = (150, 150, 150)
saucer_y = 130
cabin_y = 120
saucer_start_x = -150
cabin_start_x = -110
ship_speed = 5
```

```python

# 2
# Make a package using a rectangle
package = pygame.Rect(0, 0, 30, 30)
package_x = 385
package_start_y = 145
package_speed = 7

# 3
# Start the animating objects where they begin on screen
ship_saucer.y = saucer_y
ship_cabin.y = cabin_y
package.x = package_x
package.y = package_start_y

# 4
# Animation variables
ship_offset = 0
package_offset = 510  # the package starts off-screen the first time
packages_delivered = 0


#### ---- ANIMATION LOOP ---- ####

while packages_delivered < 3:

    #### ---- UPDATE POSITIONS ---- ####

    # 5
    # Move ship
    ship_offset += ship_speed
    ship_saucer.x = saucer_start_x + ship_offset
    ship_cabin.x = cabin_start_x + ship_offset

    # 6
    # Move package
    package_offset += package_speed
    package.y = package_start_y + package_offset

    # 7
    # Reset ship if off-screen
    if ship_saucer.x >= 950:
        ship_offset = 0
        packages_delivered += 1
```

```python

    # 8
    # Reset package if ship is halfway across (x is 325)
    if ship_saucer.x == 325:
        package_offset = 0


    #### ---- DRAW ---- ####

    # 9
    # Draw background
    window.fill((15, 5, 70))
    pygame.draw.circle(window, (45, 185, 225), (400, 900), 500)

    # 10
    # Draw package
    pygame.draw.rect(window, (50, 175, 35), package)

    # 11
    # Draw ship
    pygame.draw.ellipse(window, ship_color, ship_saucer)
    pygame.draw.ellipse(window, ship_color, ship_cabin)

    # 12
    # Finish drawing
    pygame.display.flip()
    pygame.time.wait(15)



# Turn in your Coding Exercise.


```

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3.3 Stick Dance: Smooth

Code Restructure

StickDanceSmooth_Solution.py

```
"""
LESSON: 3.3 - Animation
EXERCISE: Stick Dance (Smooth)
"""

#### ---- SETUP ---- ####

import random
import pygame
pygame.init()
window = pygame.display.set_mode([300, 500])
frames = 0


#### ---- LEG ANIMATION VARIABLES ---- ####

# Assign 25 to l_leg_start


# Assign 150 to r_leg_start


# Assign 0 to leg_offset


# Assign 4 to leg_speed



#### ---- ARM ANIMATION VARIABLES ---- ####


# Assign 100 to l_arm_start


# Assign 250 to r_arm_start


# Assign 0 to arm_offset
```

```


# Assign 3 to arm_speed



#### ---- ANIMATION LOOP ---- ####

# Increase the number of frames from 100 to 300
while frames < 100:

    frames += 1
    window.fill((255, 255, 255))


    #### ---- BODY ---- ####

    pygame.draw.circle(window, (0, 0, 0), (150, 75), 50, 1)
    pygame.draw.line(window, (0, 0, 0), (150, 125), (150, 275))


    #### ---- LEGS ---- ####

    # Instead of the random value, assign left_leg the
    # value l_leg_start plus leg_offset
    left_leg = random.randint(25, 100)

    # Instead of the random value, assign right_leg the
    # value r_leg_start plus leg_offset
    right_leg = random.randint(200, 275)

    # Increment leg_offset by leg_speed
    # ---> TEST AFTER THIS LINE <--- #


    # If leg_offset is more than 125 or less than 0


        # Multiply leg_speed by -1 and assign it to
        # leg_speed
        # ---> TEST AFTER THIS LINE <--- #

```

```

    pygame.draw.line(window, (0, 0, 0), (150, 275), (left_leg, 475))
    pygame.draw.line(window, (0, 0, 0), (150, 275), (right_leg, 475))


    #### ---- ARMS ---- ####

    # Instead of the random value, assign left_arm the
    # value l_arm_start plus arm_offset
   left_arm = random.randint(100, 250)

    # Instead of the random value, assign right_arm the
    # value r_arm_start minus arm_offset
    right_arm = random.randint(100, 250)

    # Increment arm_offset by arm_speed
    # ---> TEST AFTER THIS LINE <--- #


    # If arm_offset is more than 200


        # Assign arm_offset the value 200


        # Multiply arm_speed by -1 and assign back to
        # arm_speed
        # ---> TEST AFTER THIS LINE <--- #


    # Elif arm_offset is less than 25


        # Assign arm_offset 25


        # Multiply arm_speed by -1 again and assign
        # back to arm_speed
        # ---> TEST AFTER THIS LINE <--- #


```

```
    # Replace the two lines below with one line: draw a
    # BLACK LINE from (25, left_arm) to (275, right_arm)
    # ---> TEST AFTER THIS LINE <--- #
    pygame.draw.line(window, (0, 0, 0), (25, left_arm), (150, 175))
    pygame.draw.line(window, (0, 0, 0), (275, right_arm), (150, 175))


    #### ---- FINISH ---- ####

    pygame.display.flip()

    # Change the WAIT milliseconds from 100 to 10
    # ---> TEST AFTER THIS LINE <--- #
    pygame.time.wait(100)



# Turn in your Coding Exercise.

```

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3.3 Stick Dance: Smooth

Code Restructure

StickDanceSmooth_Solution.py

```python
"""
LESSON: 3.3 - Animation
EXERCISE: Stick Dance (Smooth)
"""

#### ---- SETUP ---- ####

import random
import pygame
pygame.init()
window = pygame.display.set_mode([300, 500])
frames = 0


#### ---- LEG ANIMATION VARIABLES ---- ####

# Assign 25 to l_leg_start
l_leg_start = 25

# Assign 150 to r_leg_start
r_leg_start = 150

# Assign 0 to leg_offset
leg_offset = 0

# Assign 4 to leg_speed
leg_speed = 4


#### ---- ARM ANIMATION VARIABLES ---- ####


# Assign 100 to l_arm_start
l_arm_start = 100

# Assign 250 to r_arm_start
r_arm_start = 250

# Assign 0 to arm_offset
```

```
arm_offset = 0

# Assign 3 to arm_speed
arm_speed = 3


#### ---- ANIMATION LOOP ---- ####

# Increase the number of frames from 100 to 300
while frames < 300:

    frames += 1
    window.fill((255, 255, 255))


    #### ---- BODY ---- ####

    pygame.draw.circle(window, (0, 0, 0), (150, 75), 50, 1)
    pygame.draw.line(window, (0, 0, 0), (150, 125), (150, 275))


    #### ---- LEGS ---- ####

    # Instead of the random value, assign left_leg the
    # value l_leg_start plus leg_offset
    left_leg = l_leg_start + leg_offset

    # Instead of the random value, assign right_leg the
    # value r_leg_start plus leg_offset
    right_leg = r_leg_start + leg_offset

    # Increment leg_offset by leg_speed
    # ---> TEST AFTER THIS LINE <--- #
    leg_offset += leg_speed

    # If leg_offset is more than 125 or less than 0
    if leg_offset > 125 or leg_offset < 0:

        # Multiply leg_speed by -1 and assign it to
        # leg_speed
        # ---> TEST AFTER THIS LINE <--- #
        leg_speed = leg_speed * -1
```

```python

    pygame.draw.line(window, (0, 0, 0), (150, 275), (left_leg, 475))
    pygame.draw.line(window, (0, 0, 0), (150, 275), (right_leg, 475))


    #### ---- ARMS ---- ####

    # Instead of the random value, assign left_arm the
    # value l_arm_start plus arm_offset
    left_arm = l_arm_start + arm_offset

    # Instead of the random value, assign right_arm the
    # value r_arm_start minus arm_offset
    right_arm = r_arm_start - arm_offset

    # Increment arm_offset by arm_speed
    # ---> TEST AFTER THIS LINE <--- #
    arm_offset += arm_speed

    # If arm_offset is more than 200
    if arm_offset > 200:

        # Assign arm_offset the value 200
        arm_offset = 200

        # Multiply arm_speed by -1 and assign back to
        # arm_speed
        # ---> TEST AFTER THIS LINE <--- #
        arm_speed = arm_speed * -1

    # Elif arm_offset is less than 25
    elif arm_offset < 25:

        # Assign arm_offset 25
        arm_offset = 25

        # Multiply arm_speed by -1 again and assign
        # back to arm_speed
        # ---> TEST AFTER THIS LINE <--- #
        arm_speed = arm_speed * -1

```

```
    # Replace the two lines below with one line: draw a
    # BLACK LINE from (25, left_arm) to (275, right_arm)
    # ---> TEST AFTER THIS LINE <--- #
    pygame.draw.line(window, (0, 0, 0), (25, left_arm), (275, right_arm))


    #### ---- FINISH ---- ####

    pygame.display.flip()

    # Change the WAIT milliseconds from 100 to 10
    # ---> TEST AFTER THIS LINE <--- #
    pygame.time.wait(10)



# Turn in your Coding Exercise.

```

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3.3 Bouncing Ball

Code Writing

BouncingBall_solution.py

```
"""
LESSON: 3.3 - Animation
EXERCISE: Bouncing Ball
"""

#### ---- SETUP ---- ####

# Import PYGAME


# INITialize pygame


# Create a new window of size [800, 300] with SET_MODE
# and assign to variable window



#### ---- ANIMATION VARIABLES ---- ####

# Set x_speed to 2


# Set y_speed to 3


# Set ball_x to 0


# Set ball_y to 0


# Set ceiling_y to 0


# Set floor_y to 285


```

```
# Set ceiling_increment to 100
# ---> TEST AFTER THIS LINE <--- #



#### ---- ANIMATION LOOP ---- ####

# While ball_x is less than 800



    #### ---- MOVE THE BALL ---- ####

    # Increment ball_x by x_speed
    # ---> TEST AFTER THIS LINE <--- #


    # Increment ball_y by y_speed



    #### ---- BOUNCE THE BALL ---- ####

    # If ball_y is greater than floor_y


        # Multiply y_speed by -1 and assign to y_speed


        # Increment ceiling_y by ceiling_increment
        # typecast to an int


        # Multiply ceiling_increment by .5 and assign
        # the result to ceiling_increment
        # ---> TEST AFTER THIS LINE <--- #


    # If ball_y is less than ceiling_y


        # Multiply y_speed by -1 and assign to y_speed
        # ---> TEST AFTER THIS LINE <--- #

```

```


    #### ---- DRAW THE BALL ---- ####

      #  FILL  the  window  with
   WHITE  )

    # Draw a CIRCLE in the window with any COLOR at
    # position (ball_x, ball_y) with radius 15


    # FLIP the display


    # WAIT for 10 milliseconds
    # ---> TEST AFTER THIS LINE <--- #



# Turn in your Coding Exercise.

```

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3.3 Bouncing Ball

Code Writing

BouncingBall_solution.py

```python
"""
LESSON: 3.3 - Animation
EXERCISE: Bouncing Ball
"""

#### ---- SETUP ---- ####

# Import PYGAME
import pygame

# INITialize pygame
pygame.init()

# Create a new window of size [800, 300] with SET_MODE
# and assign to variable window
window = pygame.display.set_mode([800, 300])


#### ---- ANIMATION VARIABLES ---- ####

# Set x_speed to 2
x_speed = 2

# Set y_speed to 3
y_speed = 3

# Set ball_x to 0
ball_x = 0

# Set ball_y to 0
ball_y = 0

# Set ceiling_y to 0
ceiling_y = 0

# Set floor_y to 285
floor_y = 285

```

```
# Set ceiling_increment to 100
# ---> TEST AFTER THIS LINE <--- #
ceiling_increment = 100


#### ---- ANIMATION LOOP ---- ####

# While ball_x is less than 800
while ball_x < 800:


    #### ---- MOVE THE BALL ---- ####

    # Increment ball_x by x_speed
    # ---> TEST AFTER THIS LINE <--- #
    ball_x += x_speed

    # Increment ball_y by y_speed
    ball_y += y_speed


    #### ---- BOUNCE THE BALL ---- ####

    # If ball_y is greater than floor_y
    if ball_y > floor_y:

        # Multiply y_speed by -1 and assign to y_speed
        y_speed *= -1

        # Increment ceiling_y by ceiling_increment
        # typecast to an int
        ceiling_y += int(ceiling_increment)

        # Multiply ceiling_increment by .5 and assign
        # the result to ceiling_increment
        # ---> TEST AFTER THIS LINE <--- #
        ceiling_increment *= .5

    # If ball_y is less than ceiling_y
    if ball_y < ceiling_y:

        # Multiply y_speed by -1 and assign to y_speed
        # ---> TEST AFTER THIS LINE <--- #
        y_speed *= -1
```

```


    #### ---- DRAW THE BALL ---- ####

    # FILL the window with WHITE
    window.fill((255, 255, 255))

    # Draw a CIRCLE in the window with any COLOR at
    # position (ball_x, ball_y) with radius 15
    pygame.draw.circle(window, (100, 100, 255), (ball_x, ball_y), 15)

    # FLIP the display
    pygame.display.flip()

    # WAIT for 10 milliseconds
    # ---> TEST AFTER THIS LINE <--- #
    pygame.time.wait(10)


# Turn in your Coding Exercise.

```

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